YIELD AND YIELD SPREAD MEASURES FOR FLOATING-RATE NOTES

Learning Outcome

- calculate and interpret yield spread measures for floating-rate instruments

Yield and Yield Spread Measures for Floating-Rate Instruments

Floating-rate instruments, including floating-rate notes (FRNs or floaters) and most loans, are different from fixed-rate bonds. Rather than fixed coupon payments, they vary from period to period depending on the current level of a reference interest rate. The intent is to both automatically adjust a borrower's base rate to market conditions and offer an investor or lender less price risk when market interest rates fluctuate. In principle, a floater has a stable price even in a period of volatile interest rates because cash flows adjust with changes in interest rates. With a traditional fixed-income security, constant future cash flows result in price changes in response to interest rate volatility.

Exhibit 1 shows the term sheet for four-year FRNs issued by Antelas AG, the German emerging technology company focused on manufacturing process design, introduced in an earlier lesson.

Exhibit 1:

Antelas AG Floating-Rate Note Issue

Antelas AG Four-Year Floating-Rate Notes (the "Notes") Prospectus Summary	
Issuer:	Antelas AG
Settlement Date:	[T + 5 Business Days]
Maturity Date:	[Four Years from Settlement Date]
Principal Amount:	EUR250 million
Interest:	MRR plus 250 bps p.a.
Interest Payment:	MRR is reset quarterly and interest is paid quarterly Commencing three months from [Settlement Date] to be paid quarterly with final payment on [Maturity Date]
Seniority:	The Notes are secured and unsubordinated obligations of Antelas AG and will rank pari passu with all other secured and unsubordinated indebtedness
Business Days:	Frankfurt

Notice how the interest is stated "MRR plus 250 bps p.a." The market reference rate (MRR) on an FRN or loan usually is a short-term money market rate. Typically, the reference rate is determined at the beginning of the period, and the interest payment is made at the end of the period. This payment structure is called "in arrears." The most common day-count conventions for calculating accrued interest on floaters are actual/360 and actual/365. For the Antelas FRN, the MRR is reset quarterly and interest is paid quarterly.

For most FRNs and loans, a specified spread is added to or subtracted from the market reference rate. For the Antelas FRN, the spread over the MRR is 250 bps, and this is

margin.

The **required margin** is the yield spread over or under the reference rate such that the FRN is priced at par value on a rate reset date. It is determined by the market. Suppose that a floater is issued at par value and pays MRR plus 0.50%. The quoted margin is 50 bps. If there is no change in the credit risk of the issuer, the required margin remains at 50 bps. On each quarterly reset date, the floater will be priced at par value. Between coupon dates, its flat price will be at a premium or discount to par value if MRR goes, respectively, down or up. However, if the required margin continues to be the same as the quoted margin, the flat price is "pulled to par" as the next reset date nears. At the reset date, any change in MRR is included in the interest payment for the next period.

Changes in the required margin usually come from changes in the issuer's credit risk. Changes in liquidity or tax status can also affect the required margin. Note that these are the same factors that affect the yield spread for fixed-rate bonds. Suppose that on a reset date, the required margin increases to 75 bps because of a downgrade in the issuer's credit rating. If the quoted margin is 50 bps, the issuer now pays its investors a "deficient" interest payment. This FRN will be priced at a discount below par value. The amount of the discount is the present value of the deficient future cash flows. That annuity is 25 bps per period for the remaining life of the bond. It is the difference between the required and quoted margins. If the required margin instead fell to 40 bps, perhaps from a decrease in the issuer's credit risk, the FRN would be priced at a premium. The amount of the premium is the present value of the 10 bp annuity for the "excess" interest payment each period.

Fixed-rate and floating-rate bonds are essentially the same with respect to changes in credit risk. With fixed-rate bonds, the premium or discount arises from a difference in the fixed coupon rate and the required yield-to-maturity. With floating-rate bonds, the premium or discount arises from a difference in the fixed quoted margin and the required margin. However, fixed-rate and floating-rate bonds are very different with respect to changes in benchmark interest rates.

The valuation of a floating-rate note requires a pricing model. Recall that the price for a fixed-rate bond given a market discount rate *r* and a coupon per period *PMT* is

$$PV = \frac{PMT}{(1+r)^1} + \frac{PMT}{(1+r)^2} + \ldots + \frac{PMT + FV}{(1+r)^N}.$$

For an FRN, *PMT* is a function of the MRR and the quoted margin and *r* is a function of the MRR and the discount margin. We can use this intuition to derive Equation 1, a simplified FRN pricing model.

$$PV = \frac{\frac{(MRR+QM)\times FV}{m}}{\left(1+\frac{MRR+DM}{m}\right)^1} + \frac{\frac{(MRR+QM)\times FV}{m}}{\left(1+\frac{MRR+DM}{m}\right)^2} + \ldots + \frac{\frac{(MRR+QM)\times FV}{m} - 1}{\left(1+\frac{MRR+DM}{m}\right)^N}$$

where

PV = present value, or the price of the floating-rate note

MRR = the market reference rate, stated as an annual percentage rate (it is sometimes known generically as Index)

QM = the quoted margin, stated as an annual percentage rate

FV = the future value paid at maturity, or the par value of the bond

m = the periodicity of the floating-rate note, the number of payment periods per year

DM = the discount margin = required margin stated as an annual percentage rate

N = the number of evenly spaced periods to maturity

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there are *N* evenly spaced periods to maturity. There is no accrued interest, so the flat price is the full price. Second, the model assumes a 30/360 day-count convention, to make sure the periodicity, *m*, is an integer. However, in practice, most floaters use actual/360 day counts. Third, and most important, the same *MRR* is used for all cash flows. More complex FRN pricing models use projected future rates in the numerators and spot rates in the denominators. Therefore, the calculation for *DM* depends on the simplifying assumptions in the pricing model.

Suppose that we are pricing a two-year, semi-annual FRN that pays MRR plus 0.50%. Assume MRR is 1.25% and the yield spread required by investors is 40 bps. For Equation 1, we have the following inputs:

MRR = 0.0125,

QM = 0.0050,

FV = 100,

m = 2,

DM = 0.0040, and

N = 4.

Using Equation 1 for *N* = 4, the FRN is priced at 100.196 per 100 of par value. This floater is priced at a premium above par value because the quoted margin is greater than the discount margin.

$$PV = \frac{\frac{(MRR+QM)\times FV}{m}}{(1+\frac{MRR+DM}{m})^{1}} + \frac{\frac{(MRR+QM)\times FV}{m}}{(1+\frac{MRR+DM}{m})^{2}} + \ldots + \frac{\frac{(MRR+QM)\times FV}{m}+FV}{(1+\frac{MRR+DM}{m})^{N}}$$

$$PV = \frac{\frac{(0.0125+0.0050)\times 100}{2}}{(1+\frac{0.0125+0.0040}{2})^{1}} + \frac{\frac{(0.0125+0.0050)\times 100}{2}}{(1+\frac{0.0125+0.0040}{2})^{2}} + \frac{\frac{(0.0125+0.0050)\times 100}{2}}{(1+\frac{0.0125+0.0040}{2})^{3}} +$$

PV = 100.196

If we have the market price of an FRN, we can also use Equation 1 to estimate the discount margin. Suppose that a two-year FRN pays MRR plus 0.75% on a semiannual basis. Currently, MRR is 1.10% and the price of the floater is 95.50 per 100 of par value, a discount to par, because of worsening credit risk. For Equation 1, we have the following inputs:

PV = 95.50,

MRR = 0.0110,

QM = 0.0075,

FV = 100,

m = 2, and

N = 4.

Using the Solver add-in for Microsoft Excel, we find *DM* = 3.12%, or 312 bps, by setting the price (sum of present values of cash flows) equal to 95.50 as the objective and *DM* as the change variable. Please see the candidate learning ecosystem online for a spreadsheet demonstrating the calculation in the downloadable Excel workbook.

If this FRN was issued at par value, investors required at that time a spread of only 75 bps over MRR. Now, after the credit downgrade, investors require an *estimated* discount margin of 312 bps. The floater trades at a discount because the quoted margin remains

banks and other fixed-income instruments based on MRR. Because a significant portion of bank funding comes from short-term deposits and other floating-rate sources, banks prefer to make floating-rate loans to businesses and individuals, as opposed to fixed-rate loans, to help maintain a match between assets and liabilities on the balance sheet.

EXAMPLE 1

Calculating the Discount Margin for a Floating-Rate Note

Suppose that the MRR for the Antelas AG four-year floating-rate note is on a three-month basis and the quoted margin is 250 bps. The floater is priced at 97 per 100 of par value. Calculate the discount margin for the floater assuming that three-month MRR is constant at –0.55%. Assume a 30/360 day-count convention and evenly spaced periods.

Solution:

We can use the Solver add-in in Microsoft Excel to quickly solve for *DM* = 3.29%, or 329 bps, in the same manner as in the prior example. Please see the candidate learning ecosystem online for a spreadsheet demonstrating the calculation in the downloadable Excel workbook.

Alternatively, we can use the RATE function in Microsoft Excel, Google Sheets, or a financial calculator by first solving for *PMT*, the numerators in Equation 1.

$$PMT = \frac{(MRR + QM) \times FV}{m}.$$

$$PMT = \frac{(-0.0055 + 0.0250) \times 100}{4}.$$

$$PMT = 0.4875.$$

Then, using *PV* = –97, *FV* = 100, *PMT* = 0.4875, and *N* = 16, we can solve for the discount rate per period of 0.686122%. We use this to solve for *DM*.

$$r = \frac{MRR + DM}{m}.$$

$$0.00686122 = \frac{-0.0055 + DM}{4}.$$

$$DM = 0.0329 = 3.29\%.$$

The quoted margin is 250 bps over the reference rate. Using the simplified FRN pricing model, the discount margin (the spread investors require for the floater to be priced at par) is 329 bps.

Question Set

1. *Fill in the blanks using one of the words in each set of parentheses:*

Floaters with longer reset periods may be _____ (more, less) exposed to interest rate and price volatility.

The _____ (shorter, longer) the reset period, the more a floater will behave similarly to a short-dated fixed-rate security and the _____ (less, more) its price will potentially fluctuate.

Hide Solution

Solution:

fluctuate.

2. For a five-year floating-rate security, if market interest rates change by 0.5%, the change in the price of the security will most likely be:

A. zero.
B. like an otherwise identical fixed-rate security.
C. related to the security's coupon reset frequency.

Hide Solution

Solution:

C is correct. The interest rate sensitivity of a floating-rate security comes typically from the time remaining until its next coupon reset date.

A is incorrect: when the interest rates changes, the discount rate of the floating rate security also changes, changing the present value of the security when discounting the cashflows.

B is incorrect: with a fixed-rate security, interest rates changes affect the price more than an identical (same-term) floating-rate security, due to its fixed coupon payments and a higher duration.

3. Identify the relationship between required margin and a floater's price at the reset date:

1. Required margin > Quoted margin	A. Floater priced at par
2. Required margin = Quoted margin	B. Floater priced at a discount
3. Required margin < Quoted margin	C. Floater priced at a premium

Hide Solution

Solution:

1. Required margin > Quoted margin	B. Floater priced at a discount
2. Required margin = Quoted margin	A. Floater priced at par
3. Required margin < Quoted margin	C. Floater priced at a premium

For Required margin > Quoted margin, B is correct. Changes in the required margin usually come from changes in the issuer's credit risk. If the required margin increases and is above the quoted margin at the reset date, the floater pays a "deficient" interest payment. The floater will be priced at a discount below par value.

For Required margin = Quoted margin, A is correct. If the required margin continues to be the same as the quoted margin, the flat price is "pulled to par" as the next reset date nears.

For Required margin < Quoted margin, C is correct. If the required margin decreases and is below the quoted margin, the floater will be priced at a premium.

4. A two-year Italian floating-rate note pays three-month Euribor of –0.50% plus 250 bp. The floater is priced at 99 per 100 of par value. Assuming the 30/360 day-count convention and evenly spaced periods, the discount margin for the floater is closest to:

A. 201 bps.
B. 251 bps.
C. 300 bps.

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information provided, the periodic *PMT* is equal to

$$\frac{(MRR + QM) \times FV}{m} = \frac{(-0.005 + 0.0025) \times 100}{4} = 0.5.$$

Using the RATE function in Microsoft Excel, Google Sheets, or a financial calculator using *PV* = –99, *FV* = 100, *PMT* = 0.5, and *N* = 8, we solve for the discount rate per period of 0.628562%. The discount margin can be estimated by solving for *DM*:

$$r = \frac{MRR + DM}{m}.$$

$$0.00628562 = \frac{-0.005 + DM}{4}.$$

$$DM = 0.03014 = 3.014\%.$$

5. The Antelas AG four-year floating-rate note pays three-month MRR of –0.55% plus 250 bps. The floater is priced at 102 per 100 of par value. Assuming the 30/360 day-count convention and evenly spaced periods, calculate the discount margin for the floating-rate note.

Hide Solution

Solution:

The estimated discount margin is 199 bps.

Using the information provided, the periodic *PMT* is equal to

$$\frac{(MRR + QM) \times FV}{m} = \frac{(-0.0055 + 0.025) \times 100}{4} = 0.4875.$$

Using the RATE function in Microsoft Excel, Google Sheets, or a financial calculator using *PV* = –102, *FV* = 100, *PMT* = 0.4875, and *N* = 16, we solve for the discount rate per period of 0.358655%. The discount margin can then be estimated by solving for *DM*:

$$r = \frac{MRR + DM}{m}.$$

$$0.00358655 = \frac{-0.0055 + DM}{4}.$$

$$DM = 0.019846 = 1.99\%.$$

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SY Solution to Question 4 is incorrect. 250bps should be 0.0250 not 0.0025. This calculation is wrong... No correct solution exists.

Created 9 months ago by Samuel Yingxu Chan 6 replies | Last Activity: 2 months ago

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SV A two-year floating-rate note issued by a French corporation pays the three-month MRR of –0.55% plus 160 bps. The floater is priced at 101.20 per 100 of par value. Assuming the 30/360 day-count convention and 90 days per period, the discount margin for the floater is closest to:

A.25 bps.

B.50 bps.

C.110 bps.

For the given question, isn't the answer option C? But the answer in practice dashboard in the CFA curriculum is given as Option b) 50. How is

Created 4 months ago by Sanju Vikashini Arumugam

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